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RESEARCH ARTICLE

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Twenty-eight days of repeated dose sub-acute toxicological evaluation of polyherbal Ayurvedic medicine BPGrit in Sprague–Dawley rats

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Abstract

A pre-clinical toxicological evaluation of herbal medicines is necessary to identify any underlying health-associated side effects, if any. BPGrit is an Ayurveda-based medicine prescribed for treating hypertensive conditions. High-performance liquid chromatography-based analysis revealed the presence of gallic acid, ellagic acid, coumarin, cinnamic acid, guggulsterone E, and guggulsterone Z in BPGrit. For sub-acute toxicity analysis of BPGrit, male and female Sprague–Dawley rats were given repeated oral gavage at 100, 300, and 1000 mg/kg body weight/day dosages for 28 days, followed by a 14-day recovery phase. No incidences of mortality, morbidity, or abnormal clinical signs were observed in BPGrit-treated rats throughout the study period. Also, the body weight and food consumption habits of the experimental animals did not change during the study duration. Hematological, biochemical, and histopathological analysis did not indicate any abnormal changes occurring in the BPGrit-treated rats up to the highest tested dose of 1000 mg/kg body weight/day. Finally, the study established the “no-observed-adverse-effect level” for BPGrit at >1000 mg/kg body weight/day in Sprague–Dawley rats.

KEYWORDS

14-day recovery period, 28-day repeated dose toxicity, Ayurveda, BPGrit, Sprague–Dawley rats

1 | INTRODUCTION

The World Health Organization (WHO, 2022) estimates that approximately 80% of the world population uses herbal medicines for treating various diseases, and 170 of the 194 member states have documented them as traditional medicines. These herbal remedies may comprise of monoherbal or polyherbal components in crude or processed forms (Calixto, 2000). Therapeutic characteristics of these herbal medicines originate from their active phytochemical components, which may have subtle, immediate, or latent health effects (Jităreanu et al., 2023; Wang et al., 2021; Woo et al., 2012). WHO (2004) has developed pharmacovigilance guidelines that recommend that each member state frame and implement a regulatory structure for controlling the appropriate manufacturing and therapeutic use of traditional medicine. Internationally, regulatory safety guidelines for pharmaceutical and herbal medicines are enforced by regulatory agencies like “Organisation for Economic Co-operation and Development” (OECD, 2024) and “International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use” that provide standardized safety testing guidelines (ICH, 2024).

Ayurveda is an alternative form of medicine that originated in India. The system has been well documented through ancient textbooks, and the term “Ayurveda” means “The Science of Life” (Agnivesha, 1994; Baidyanath, 2021). Generally, herbal remedies prescribed in Ayurveda are considered safe for human health due to their clinical usage over millennia. In India, Ayurvedic formulations are regulated under the revised Drug and Cosmetics Act, 1940, and regulations of 1945 (Ministry of AYUSH, 2023). In the United States, it is considered a complementary and alternative medicine (CAM) and referred to as medical and health care systems, practices, and products that are not part of modern medicine practices. According to US Food and Drug Administration (US FDA) rules, if used to treat illnesses, CAMs might be deemed "drugs." Such CAMs would have to undergo rigorous pre-clinical and clinical testing, much like their allopathic equivalents, in accordance with the US FDA (2007) Food, Drug, and Cosmetic Act ("FD&C Act") and/or the Public Health Services Act.

In India, the Central Council for Research in Ayurvedic Sciences (CCRAS, 2018), functioning under the Ministry of AYUSH, Government of India, has issued comprehensive guidelines for assessing the oral acute, sub-acute, and chronic toxicity of herbal medicines in experimental animals. These directives are adapted from the internationally well-established OECD-based Good Laboratory Practice guidelines with minor modifications. The OECD 407 guideline has been extensively used for evaluating the oral toxicity, and "no-observed-adverse-effect level" (NOAEL) of herbal formulations and their phytochemicals (Amroun et al., 2021; Balkrishna et al., 2022; Kumar et al., 2015; OECD, 2008; Rehman et al., 2022; Variya et al., 2019).

BPGrit is a polyherbal Ayurveda-based medicine prescribed for treating hypertensive conditions. BPGrit has been prepared using plant extracts obtained from *Commiphora mukul* (Hook. Ex Stocks) Engl., *Terminalia arjuna* (Roxb. ex DC.) Wight & Arn., *Tribulus terrestris* L., *Punica granatum* L., *Allium sativum* L., and *Cinnamomum zeylanicum* Blume. Extracts obtained from *C. mukul* (Hook. Ex Stocks) Engl. contains 38% oleo-gum resin, which has been linked to the development of skin rashes, irregular menstruation, intestinal discomfort, and hepatotoxicity (National Toxicology, 2020; Sarup et al., 2015). *T. arjuna* (Roxb. Ex DC.) Wright & Arn. has been reported to have no side effects after consuming a single and repeated dose of 2000 mg/kg body weight (Thomas & Radhakrishnan, 2023). In an in vivo acute toxicity study conducted in Wistar rats, *T. terrestris* L. methanolic extract was determined to be non-toxic up to the tested concentration of 3000 mg/kg body weight (Abbas et al., 2022). *P. granatum* L. fruit extract was found to be non-toxic up to the oral dose of 600 mg/kg body weight/day performed over a 90-day period in Wistar rats (Patel et al., 2008). In a 4-week-long sub-chronic study done in male and female Wistar rats, *A. sativum* L. was found to induce liver and lung injuries at high doses of 1200 and 4800 mg/kg body weight (Gatsing et al., 2006). Plant extracts obtained from *C. zeylanicum* Blume were found to be non-toxic and non-reactive in the African green monkey kidney (Vero) cells (Alanazi & Almohammed, 2022).

In this study, we explored the repeated toxicity of BPGrit in male and female Sprague–Dawley rats over a 28-day treatment regimen using oral gavage, followed by a 14-day recovery phase to assess any potential reversal or delayed responses following OECD 407 guidelines. BPGrit was administered at doses of 0, 100, 300, and 1000 mg/kg body weight/day to the treatment group animals. Animals in the treatment group were euthanized on Day 29 following the completion of their designated treatment regimen, while those in the recovery group (0 and 1000 mg/kg body weight/day) were euthanized 14 days thereafter. The evaluation encompassed various parameters, including mortality, clinical signs, detailed clinical examination, changes in body weight, feed consumption, ophthalmoscopy examination, functional observation, hematology, clinical chemistry, urine analysis, organ weights, and gross and histopathological examination, conducted during both the treatment and recovery periods. The observed changes in mortality, morbidity, and clinical biomarkers were utilized to determine the NOAEL for BPGrit in Sprague–Dawley rats.

2 | MATERIAL AND METHODS

2.1 | Test substance

BPGrit (Batch Number: D4/CHM/ANCA-190/1121) was sourced from Divya Pharmacy (Uttarakhand, India) and stored at room temperature according to the manufacturer's instructions.

2.2 | Reagents

Methyl cellulose was obtained from Loba Chemie, Maharashtra, India. Harris-modified hematoxylin, eosin stains, and standard phosphate-buffered saline solution were purchased from Sigma-Aldrich, MO, USA. Finar, Gujarat, India, provided ultra-high-performance liquid chromatography (UHPLC) quality acetonitrile. Rankem, Maharashtra, India, supplied methanol, orthophosphoric acid analytical reagent (AR) grade, and diethylamine. Tropicamide was obtained from Akums Drug and Pharmaceuticals Ltd., Uttarakhand, India. A Milli-Q system (Millipore, MA, USA) was used to get deionized water. Gallic acid (potency: 97.9%), ellagic acid (potency: 99.6%), and coumarin (potency: 98.0%) were purchased from Sigma-Aldrich. Cinnamic acid (potency: 99.7%) was purchased from Sisco Research Laboratories (SRL) Pvt. Ltd., Maharashtra, India, and guggulsterone E (potency: 98.1%) and guggulsterone Z (potency: 97.6%) were purchased from the source Natural Remedies, Uttar Pradesh, India.

2.3 | High-performance liquid chromatography (HPLC) analysis

Standards for gallic acid, ellagic acid, coumarin, cinnamic acid, guggulsterone E, and guggulsterone Z were individually dissolved in methanol to prepare 1 mg/mL standard stock solutions. For the analysis, 50 μg/mL of standard mix working solution was prepared from each standard stock solution. Separately, a 50 mg/mL powdered BPGrit tablet sample was dissolved in a methanol:water (90:10) solution. This mixture was centrifuged and passed through a 0.45-μm nylon syringe filter. The filtrate was used for analysis on an HPLC (Prominence-iLC-2030c 3D Plus, Shimadzu, Japan) instrument, and separation was achieved using a Shodex C18-4E (5 μm, 4.6 × 250 mm) column subjected to binary gradient elution. A gradient solvent system of solvent A (0.1% orthophosphoric acid in water) and solvent B (acetonitrile) was used. Gradient program for mobile phase B was adjusted at 0%–5% from 0 to 10 min, 5%–10% from 10 to 20 min, 10%–15% from 20 to 30 min, 15%–40% from 30 to 40 min, 40%–60% from 40 to 45 min, 60%–80% from 45 to 50 min, 80%–90% from 50 to 60 min, 90%–0% from 60 to 65 min, and 0% from 65 to 70 min. Chromatographic detection of all the analytes was performed using a photo diode array (PDA) detector at 227 nm. The temperature of the column was maintained at 27°C, and the sample injection volume was 20 μL.

2.4 | Animal housing and maintenance

For this study, 32 male Sprague–Dawley rats with body weights ranging between 178.76 and 226.54 g, and 32 nulliparous and non-pregnant female rats with body weights ranging between 151.72 and 191.21 g were purchased from Vivo Bio Tech Ltd., Telangana, India (Committee for Control and Supervision of Experiments on Animals [CCSEA], Government of India, Registration Number: 1117/PO/RcBiBt/S/07/CCSEA). All the procured animals were within the age range of 6–7 weeks upon arrival at the testing facility. Upon their arrival, the animals were quarantined for a week and given a detailed clinical examination. The animals were kept in polycarbonate cages with dimensions of 421 mm (length) × 290 mm (breadth) × 190 mm (height) in groups of three animals per cage. The cage rotation was done on a weekly basis. The temperature and humidity of the animal house were maintained at 22 ± 3°C and 30%–70%, respectively. The photoperiod of the holding area was automatically maintained at 12 h of light and dark cycles and 10–15 air changes/h throughout the study period.

Animals were fed *ad libitum* with a gamma-irradiated standard pelleted laboratory animal diet (Hylasco Biotechnology Pvt. Ltd., Telangana, India). Animals were provided with ultraviolet light and reverse osmosis (RO) sterilized drinking water *ad libitum* in autoclaved polypropylene bottles. A certified veterinarian kept close monitoring of the animals' health conditions throughout the study period. The study was carried out in accordance with OECD 407 guidelines after approval of the study protocol (Approval Number: PRIAS/LAF/IAEC-115) by the Institution Animal Ethical Committee (IAEC), Patanjali Research Institute, Uttarakhand, India.

2.5 | Study design, formulation preparation, and dosing

Male and female animals were randomly allocated to six individual groups, consisting of four main groups (G1, G2, G3, and G4) and two recovery groups (G1-R and G4-R). Each group consisted of six animals from both sexes. Methyl cellulose solution (0.5% w/v) prepared in sterilized water was used as an oral delivery vehicle in the study. Every day, the BPGrit tablets were triturated using a mortar and pestle and mixed with a freshly prepared 0.5% w/v methyl cellulose solution just before animal oral gavage administration. The stability of the BPGrit-based methyl cellulose solution was checked using the HPLC technique previously mentioned under Section 2.3 over a period of 4 h. One gram of BPGrit and 0.5% w/v methyl cellulose solution were diluted with 5 mL of methanol:water (90:10) and vortexed for 10 min. This solution was centrifuged at 10,000 rpm for 5 min and filtered using a 0.45-μm nylon syringe filter. No change was observed in the phytochemical composition of the BPGrit-based methyl cellulose solution (Figure S1 and Table S1). Animals in the G1 and G1-R (control) treatment groups received 0.5% w/v methyl cellulose vehicle control solution alone. BPGrit was given through oral gavage to the animals for 28 consecutive days at a daily dose of 100 (G2), 300 (G3), and 1000 (G4 and G4-R) mg/kg of body weight. Dosages for BPGrit were selected following the OECD 407 guideline, which recommends an upper limit of either 1000 mg/kg body weight/day, or the maximum feasible dose, whichever is higher (OECD, 2008). In the recovery (G4-R) group animals, BPGrit dosing was discontinued after 28 days, and the animals were kept under observation for an additional 14 days along with the control (G1-R) animal group.

2.6 | Observation of mortality, morbidity, clinical signs, and symptoms

The body weights of the animals were recorded individually for the G1, G2, G3, G4, G1-R, and G4-R group animals on 1, 8, 15, 22, and 28 days, and additionally on 35 and 42 days for the G1-R and G4-R group animals. Fasting body weights were taken prior to randomization and on the day of necropsy. Weekly feed consumption of all the animal groups was recorded during the study period. During the acclimation and study periods, animals were checked at least once a day for abnormal clinical signs and twice a day for morbidity and mortality. A weekly, detailed clinical examination was performed on all animals. The animals were examined for skin, fur, and eye color changes, mucous membrane secretions, excretions, and basic autonomic activity observations (e.g., lacrimation, piloerection, pupil size, and unusual respiratory pattern). Changes in the animals' gait, posture, handling reactivity, clonic or tonic movements, stereotypies, and behavior were recorded for each animal.

2.7 | Functional investigation

A battery of functional observations was carried out in all animals during the fourth week of the study and during the sixth week in the recovery group animals. The observations made were as follows:

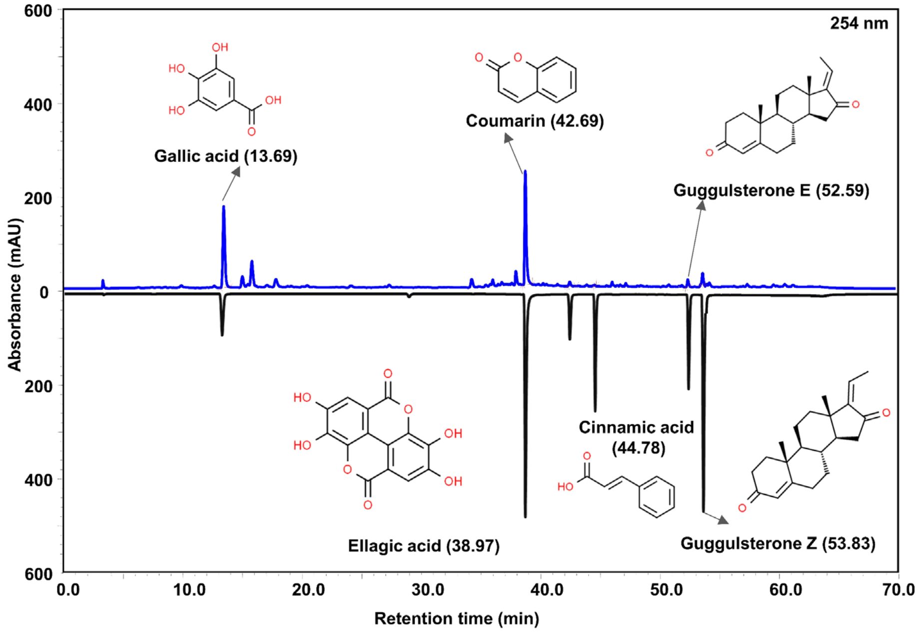

FIGURE 1 High-performance liquid chromatography (HPLC) analysis: HPLC analysis of the BPGrit medicine showed the presence of gallic acid (13.69 min), ellagic acid (38.97 min), coumarin (42.69 min), cinnamic acid (44.78 min), guggulsterone E (52.59 min), and guggulsterone Z (53.83 min).

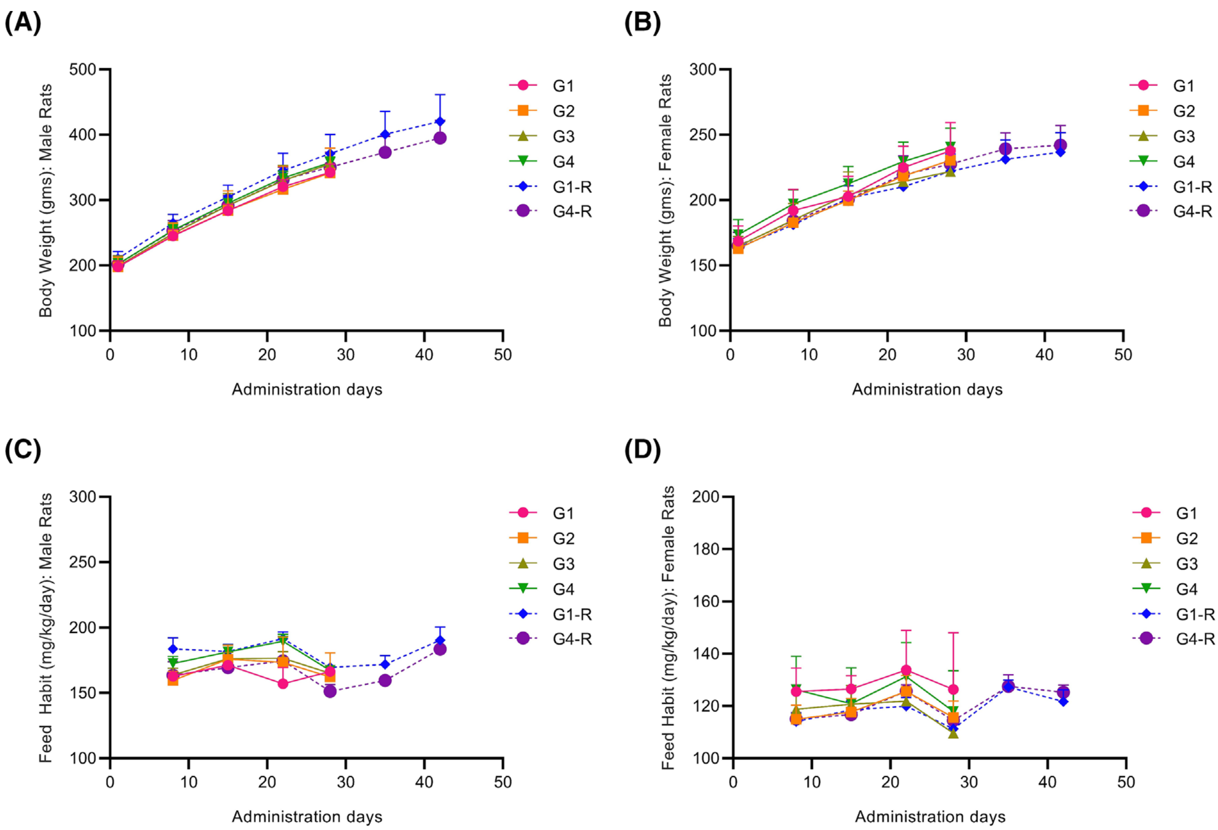

FIGURE 2 (A–D) Body weight and feed habit analysis: Body weights of the animals were recorded individually for the treatment group control (G1), 100 mg/kg body weight (G2), 300 mg/kg body weight (G3), 1000 mg/kg body weight (G4), recovery group control (G1-R), and 1000 mg/kg body weight (G4-R) animals on days 1, 8, 15, 22, and 28, and additionally on 35 and 42 days for G1-R and G4-R animals. Fasting body weights were taken prior to randomization and on the day of necropsy. Weekly feed consumption of all the animal groups was recorded during the study period. Results have been presented as mean ± SD. Statistical analysis was done using a two-way analysis of variance (ANOVA) test followed by Tukey's multiple comparison post-,hoc test.

A. **Motor activity:** Animals were acclimatized to the testing room and the activity chamber (450 mm [length] × 450 mm [breadth] × 350 mm [height]) prior to the analysis for 60 min on four consecutive days before the actual test. On the day of the examination, the animals were taken into the testing room and maintained in their home cages for 15 min before being transferred to the activity chamber for 5 min. The total number of movements done by the animals within the activity chamber was recorded automatically using an infrared actimeter (Model: LE8825, Panlab, Spain), and the number of rearings was manually tallied. The activity chamber was cleaned between individual animal studies using a damp cloth and tissue paper.

B. **Behavioral response:** Posture, ease of removal, ease of handling, alertness, mobility, fur appearance, startle, approach, and touch responses were recorded for each animal on a regular basis.

C. **Neurologic response:** Convulsions and tremors, muscle tone, gait, stereotypic movements, tail pinch response, eye blink response, landing foot splay, and air righting reflex were recorded for each animal. For the foot splay study, animals' rear feet were painted with ink. The animals were dropped on a white paper sheet from a height of around 30 cm to provide imprints during landing. The distance between the inner surfaces of each foot's fourth digit at landing was measured using the animals' imprints. The average distance was obtained after three consecutive trials for each animal.

D. **Autonomic response:** Palpebral closure, fecal consistency, urine output, lacrimation, salivation, pupil function in response to light, eye color, skin color, piloerection, ease, and rate of respiration were recorded at regular intervals.

TABLE 1 Hematological and coagulation parameters (male).

	Groups					
	G1 (vehicle control)	G2 (100 mg/kg body weight/day)	G3 (300 mg/kg body weight/day)	G4 (1000 mg/kg body weight/day)	G1-R (vehicle control)	G4-R (1000 mg/kg body weight/day)
Red blood cells ($10^6/\mu L$)	8.13 ± 0.26	8.36 ± 0.38	8.23 ± 0.26	8.00 ± 0.29	8.36 ± 0.21	8.31 ± 0.21
Hemoglobin concentration (g/dL)	16.22 ± 0.64	16.48 ± 0.70	15.94 ± 0.30	15.86 ± 0.48	15.90 ± 0.39	16.08 ± 0.51
Hematocrit (%)	44.34 ± 1.51	44.58 ± 2.24	43.86 ± 0.78	43.12 ± 1.33	42.82 ± 0.70	42.94 ± 1.68
Mean corpuscular volume (fL)	54.54 ± 1.16	53.34 ± 0.96	53.34 ± 1.29	53.94 ± 1.79	51.24 ± 0.90	51.66 ± 1.25
Mean corpuscular hemoglobin (pg)	19.96 ± 0.57	19.70 ± 0.27	19.38 ± 0.49	19.84 ± 0.75	19.04 ± 0.29	19.32 ± 0.36
Mean corpuscular hemoglobin concentration (g/dL)	36.58 ± 0.35	36.98 ± 0.33	36.36 ± 0.44	36.80 ± 0.35	37.18 ± 0.54	37.42 ± 0.34
Platelet ($10^3/\mu L$)	856.40 ± 227.77	850.20 ± 228.70	1101.80 ± 118.17	934.60 ± 216.47	965.80 ± 157.89	1017.80 ± 62.04
Reticulocyte count (%)	2.46 ± 0.45	2.90 ± 0.47	2.74 ± 0.36	2.78 ± 0.26	2.78 ± 0.52	2.62 ± 0.40
White blood cells ($10^3/\mu L$)	9.88 ± 3.98	11.17 ± 0.36	10.14 ± 2.36	11.44 ± 1.37	9.58 ± 0.94	11.16 ± 0.78*
Neutrophils (%)	8.80 ± 0.84	10.40 ± 4.28	10.60 ± 3.65	13.40 ± 1.14	8.20 ± 1.92	10.20 ± 1.92
Lymphocytes (%)	88.00 ± 1.41	87.20 ± 4.15	87.20 ± 2.68	83.40 ± 1.34*	88.60 ± 2.30	88.00 ± 1.58
Monocytes (%)	2.60 ± 0.89	1.80 ± 1.30	2.20 ± 1.10	1.80 ± 1.30	2.00 ± 0.71	1.20 ± 0.45
Eosinophils (%)	0.60 ± 0.89	0.60 ± 0.55	0.00 ± 0.00	1.40 ± 0.89	1.20 ± 0.84	0.60 ± 0.55
Basophils (%)	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Prothrombin time test (s)	18.96 ± 0.92	16.30 ± 2.92	16.68 ± 1.64	16.48 ± 1.48*	16.26 ± 1.74	17.16 ± 1.95
Activated partial thromboplastin clotting time (s)	21.50 ± 2.50	21.80 ± 2.80	20.16 ± 1.89	22.02 ± 8.57	17.68 ± 2.06	19.04 ± 1.68

Note: The number of animals per group (n) is 5. Data are presented as mean ± SD; statistical analysis was performed using one-way ANOVA followed by Dunnett's multiple comparison test.
*p-value < 0.05 (control [G1] vs. individual treatments [G2–G4], and G1-R vs. G4-R).

2.8 | Ophthalmological and clinical examinations

Ophthalmoscopic examination of treatment (G1 and G4) and recovery (G1-R and G4-R) animal groups was performed using an ophthalmoscope (Model: M2N13000, HEINE, Germany). To induce mydriasis, a few drops of 1% tropicamide were instilled in the eyes of the animals about 20 min prior to examination. Ophthalmoscopic examination of the treatment group animals was performed during the fourth week of the study, and for the recovery group animals during the sixth week of the study.

2.9 | Clinical pathology observations

Blood samples were drawn from the overnight-fasted animals under light anesthesia from the retro-orbital sinus. Blood was collected from treatment group animals on Day 29 and from recovery group animals on Day 43. Fresh blood was collected in a 10% ethylenediaminetetraacetic acid (EDTA)-containing vial, a 3.2% sodium citrate-containing vial, and a plain (without anticoagulant) vial for hematological and biochemical analysis. Hematological parameters such as whole blood samples: total erythrocyte count (10^6/μL), hematocrit %, mean corpuscular volume (fL), hemoglobin concentration (g/dL), mean corpuscular hemoglobin (pg), mean corpuscular hemoglobin concentration (g/dL), platelet count (10^3/μL), and total leukocyte count (10^3/μL) were analyzed using a hematology analyzer (Model: BC-5000 Vet, Mindray, China). Differential leucocytes were counted following Giemsa staining using the AxioScope-A1 brightfield microscope (Carl Zeiss IQS Deutschland GmbH, Oberkochen, Germany). Reticulocytes were counted in the EDTA-treated blood sample following methylene blue staining using the AxioScope-A1 brightfield microscope. For the blood coagulation assay, blood collected in 3.2% w/v sodium citrate-

TABLE 2 Hematological and coagulation parameters (female).

	Groups					
	G1 (vehicle control)	G2 (100 mg/kg body weight/day)	G3 (300 mg/kg body weight/day)	G4 (1000 mg/kg body weight/day)	G1-R (vehicle control)	G4-R (1000 mg/kg body weight/day)
Red blood cells (10^6/μL)	7.67 ± 0.40	7.74 ± 0.57	8.07 ± 0.84	8.13 ± 0.30	8.13 ± 0.33	8.46 ± 0.17
Hemoglobin concentration (g/dL)	15.32 ± 0.63	15.26 ± 1.07	15.54 ± 1.47	15.88 ± 0.63	15.74 ± 0.13	15.96 ± 0.44
Hematocrit (%)	41.34 ± 1.76	41.18 ± 2.92	42.24 ± 4.17	42.86 ± 1.40	42.26 ± 0.50	42.90 ± 1.17
Mean corpuscular volume (fL)	53.88 ± 0.50	53.24 ± 1.84	52.38 ± 1.30	52.74 ± 1.36	52.02 ± 1.89	50.68 ± 1.05
Mean corpuscular hemoglobin (pg)	19.96 ± 0.31	19.76 ± 0.61	19.26 ± 0.52	19.56 ± 0.70	19.42 ± 0.83	18.88 ± 0.40
Mean corpuscular hemoglobin concentration (g/dL)	37.02 ± 0.29	37.06 ± 0.21	36.78 ± 0.36	37.06 ± 0.43	37.30 ± 0.41	37.22 ± 0.22
Platelet (10^3/μL)	950.20 ± 174.12	918.60 ± 101.86	952.80 ± 178.72	944.60 ± 132.68	973.40 ± 112.93	1025.00 ± 120.26
Reticulocyte count (%)	2.48 ± 0.37	2.82 ± 0.42	2.80 ± 0.42	2.80 ± 0.27	2.68 ± 0.55	2.78 ± 0.31
White blood cells (10^3/μL)	6.31 ± 0.96	6.34 ± 1.57	7.50 ± 1.63	9.08 ± 0.74*	8.55 ± 1.37	7.50 ± 0.88
Neutrophils (%)	11.20 ± 2.39	8.00 ± 2.35	10.00 ± 3.00	11.80 ± 3.56	8.80 ± 1.64	8.80 ± 1.30
Lymphocytes (%)	85.20 ± 3.27	87.60 ± 4.10	85.00 ± 3.39	85.40 ± 3.78	88.40 ± 1.67	90.00 ± 1.22
Monocytes (%)	2.00 ± 1.00	2.00 ± 1.22	3.00 ± 1.22	1.20 ± 1.10	1.80 ± 0.45	0.60 ± 0.55
Eosinophils (%)	1.60 ± 1.14	2.40 ± 1.34	1.80 ± 0.84	1.60 ± 1.14	0.80 ± 0.45	0.60 ± 0.55
Basophils (%)	0.00 ± 0.00	0.00 ± 0.00	0.20 ± 0.45	0.00 ± 0.00	0.20 ± 0.45	0.00 ± 0.00
Prothrombin time test (s)	14.90 ± 1.56	14.32 ± 0.91	13.72 ± 1.29	14.02 ± 0.71	14.96 ± 1.17	20.52 ± 9.19
Activated partial thromboplastin clotting time (s)	18.80 ± 1.26	18.90 ± 2.12	19.14 ± 1.76	21.78 ± 4.94	13.60 ± 0.72	18.32 ± 2.07

Note: The number of animals per group (*n*) is 5. Data are presented as mean ± SD; statistical analysis was performed using one-way ANOVA followed by Dunnett's multiple comparison test.

*p-value < 0.05 (control [G1] vs. individual treatments [G2–G4], and G1-R vs. G4-R).

containing vials was centrifuged (Sorvall™ Legend™ Micro 21R centrifuge, Thermo Scientific, MA, USA) at 3000× g for 10 min at 4°C for isolating plasma. Prothrombin time (seconds) and activated partial thromboplastin time (seconds) were analyzed in plasma using the ECL-105 Coagulation Analyzer (Erba, Mannheim, Germany). Blood without anticoagulant was left undisturbed at room temperature for 1 h and then centrifuged at 3000× g for 15 min at 4°C (Sorvall™ Legend™ micro 21R centrifuge) for separation of serum. Aspartate transaminase (U/L), alanine transaminase (ALT) (U/L), alkaline phosphatase (ALP) (U/L), total bilirubin (mg/dL), blood urea level (mg/dL), calculated blood urea nitrogen (mg/dL), creatinine (mg/dL), glucose (mg/dL), total cholesterol (mg/dL), total protein (g/dL), albumin (g/dL), protein:albumin ratio, sodium (mmol/L), and potassium (mmol/L) were analyzed in the serum samples using the Rx Monaco system (Randox, UK).

2.10 | Urine analysis

Treatment and recovery animal groups were placed in metabolic cages on days 28 and 42, respectively. During this period, the animals were fasted and provided water *ad libitum*. Urine was collected in thymol blue crystals containing beakers and subsequently analyzed for appearance (color and clarity), volume, blood (BLD), bilirubin (BIL), urobilinogen (UBG), ketone (KET), nitrite (NIT), leucocytes (LEU), glucose (GLU), protein (PRO), pH, and specific gravity (SG) using a Laura urine analyzer (Erba).

2.11 | Necropsy and gross pathology

Treatment and recovery group animals were euthanized using thiopentone overdose on Days 29 and 43, respectively. A gross inspection of the exterior body surface, orifices, cranial, thoracic, and abdominal cavities, and their contents, was performed on all the sacrificed animals. Organs were harvested and cleaned, and adherent fatty tissues were carefully trimmed for the gross pathological examinations. Wet weights of the liver, kidneys, adrenals, testes, ovaries, epididymis, uterus, cervix, prostate, and seminal vesicles with coagulating glands, thymus, spleen, brain, and heart were recorded, and their relative organ weights were calculated in relation to terminal body weight using Equation (1):

TABLE 3 Clinical chemistry parameters (male).

	Groups					
	G1 (vehicle control)	G2 (100 mg/kg body weight/day)	G3 (300 mg/kg body weight/day)	G4 (1000 mg/kg body weight/day)	G1-R (vehicle control)	G4-R (1000 mg/kg body weight/day)
Aspartate transaminase (U/L)	128.98 ± 8.55	135.60 ± 34.95	126.78 ± 10.10	125.62 ± 13.28	136.42 ± 22.26	149.26 ± 33.06
Alanine transaminase (U/L)	50.64 ± 1.92	48.38 ± 4.91	41.38 ± 2.90*	41.30 ± 1.80*	43.72 ± 2.81	51.66 ± 7.44
Alkaline phosphatase (U/L)	261.40 ± 47.64	230.00 ± 15.41	231.40 ± 11.35	221.20 ± 32.31	150.40 ± 30.66	176.80 ± 24.29
Total bilirubin (mg/dL)	0.06 ± 0.01	0.07 ± 0.02	0.07 ± 0.02	0.07 ± 0.02	0.07 ± 0.01	0.08 ± 0.00
Blood urea level (mg/dL)	40.32 ± 4.64	47.18 ± 8.21	40.50 ± 4.57	45.98 ± 14.75	38.02 ± 5.69	41.04 ± 5.09
Calculated blood urea nitrogen (mg/dL)	18.84 ± 2.17	22.05 ± 3.83	18.93 ± 2.14	21.49 ± 6.89	17.77 ± 2.66	19.18 ± 2.37
Creatinine (mg/dL)	0.29 ± 0.05	0.31 ± 0.05	0.27 ± 0.02	0.27 ± 0.04	0.42 ± 0.03	0.35 ± 0.05*
Glucose (mg/dL)	127.88 ± 13.87	134.34 ± 7.00	118.44 ± 12.50	132.84 ± 19.71	125.06 ± 16.60	153.18 ± 16.37*
Cholesterol (mg/dL)	85.00 ± 10.25	87.00 ± 29.86	91.40 ± 9.66	80.80 ± 15.51	79.60 ± 11.95	72.60 ± 8.50
Protein (g/dL)	6.53 ± 0.22	6.45 ± 0.48	6.28 ± 0.27	6.33 ± 0.10	6.57 ± 0.26	6.63 ± 0.47
Albumin (mg/dL)	2.86 ± 0.10	2.77 ± 0.07	2.78 ± 0.10	2.78 ± 0.04	2.76 ± 0.13	2.75 ± 0.20
Protein:albumin ratio	2.29 ± 0.04	2.33 ± 0.15	2.26 ± 0.07	2.28 ± 0.02	2.38 ± 0.07	2.41 ± 0.06
Sodium (mmol/L)	146.50 ± 3.66	141.32 ± 5.34	143.70 ± 1.77	145.76 ± 3.97	151.38 ± 4.93	145.36 ± 4.38
Potassium (mmol/L)	4.88 ± 0.35	4.44 ± 0.21	4.76 ± 0.21	4.68 ± 0.23	5.09 ± 0.26	5.06 ± 0.47

Note: The number of animals per group (n) is 5. Data are presented as mean ± SD; statistical analysis was performed using one-way ANOVA followed by Dunnett's multiple comparison test.
*p-value < 0.05 (control [G1] vs. individual treatments [G2–G4], and G1-R vs. G4-R).

$$\text{Relative organ weights (g)} = \frac{\text{Organ weight (g)}}{\text{Terminal body weight (g)}} \times 100 \quad (1)$$

2.12 | Histopathology

The brain, heart, lungs, liver, kidneys, pituitary glands, thyroid, spleen, bone marrow, pancreas, adrenal gland, urinary bladder, colon, skeletal muscles, thymus, spleen, and sex organs were harvested from the individual euthanized G1 and G4 animals and histopathologically analyzed following OECD 407 guidelines (OECD, 2008). All the organs were trimmed of their adherent tissues, fixed in a 10% neutral buffered formalin solution, and embedded in paraffin wax. Using a steel-blade microtome (RM 2245, Leica Biosystems Nussloch GmbH, Nussloch, Germany), 5-µm-thick tissue sections were dissected. Microtomed tissue sections were placed on glass slides and stained with hematoxylin–eosin stain. Stained tissues were analyzed by a veterinarian histopathologist blindly using the AxioScope-A1 bright-field microscope, and representative images were acquired using an attached digital camera. Images were processed using Zeiss Zen software.

2.13 | Statistical analysis

Results were presented as mean ± standard deviation (SD) for all animal groups. Statistical analysis was performed using GraphPad Prism software (Version 7.04) (GraphPad, CA, USA). Body weight and feed consumption results were analyzed using a two-way analysis of variance (ANOVA) test followed by Tukey's multiple comparison post-hoc test. Motor activity, landing foot splay, hematology, clinical chemistry, and organ weight data were analyzed using a one-way ANOVA test followed by Dunnett's multiple comparison post-hoc test for treatment groups, and Student's paired two-tailed *t*-test for recovery groups. *p*-values < 0.05 were considered statistically significant.

TABLE 4 Clinical chemistry parameters (female).

	Groups					
	G1 (vehicle control)	G2 (100 mg/kg body weight/day)	G3 (300 mg/kg body weight/day)	G4 (1000 mg/kg body weight/day)	G1-R (vehicle control)	G4-R (1000 mg/kg body weight/day)
Aspartate transaminase (U/L)	134.98 ± 17.38	124.22 ± 13.63	115.00 ± 11.61	125.18 ± 16.15	154.98 ± 16.51	169.80 ± 8.03
Alanine transaminase (U/L)	34.04 ± 2.46	36.98 ± 5.00	34.04 ± 6.49	41.88 ± 6.72	40.94 ± 7.18	38.64 ± 4.14
Alkaline phosphatase (U/L)	125.00 ± 29.21	123.00 ± 26.11	121.22 ± 31.04	148.00 ± 28.29	129.80 ± 22.08	120.00 ± 30.11
Total bilirubin (mg/dL)	0.07 ± 0.01	0.07 ± 0.01	0.08 ± 0.01	0.07 ± 0.01	0.11 ± 0.02	0.08 ± 0.01*
Blood urea level (mg/dL)	39.18 ± 4.44	48.74 ± 8.93	44.40 ± 8.43	48.70 ± 5.11	41.42 ± 2.54	45.96 ± 5.33
Calculated blood urea nitrogen (mg/dL)	18.31 ± 2.08	22.78 ± 4.17	20.75 ± 3.94	22.76 ± 2.39	19.36 ± 1.19	21.48 ± 2.49
Creatinine (mg/dL)	0.37 ± 0.01	0.37 ± 0.01	0.37 ± 0.04	0.36 ± 0.06	0.47 ± 0.03	0.45 ± 0.03
Glucose (mg/dL)	119.98 ± 17.48	121.28 ± 9.62	108.34 ± 20.31	130.90 ± 26.11	111.36 ± 10.78	115.88 ± 18.14
Cholesterol (mg/dL)	100.20 ± 9.60	93.80 ± 10.83	101.40 ± 10.97	101.40 ± 9.91	101.20 ± 12.11	98.40 ± 6.27
Protein (g/dL)	6.19 ± 0.15	5.91 ± 0.22	6.05 ± 0.27	6.05 ± 0.40	6.42 ± 0.11	6.24 ± 0.30
Albumin (mg/dL)	2.81 ± 0.05	2.63 ± 0.19	2.75 ± 0.10	2.74 ± 0.16	2.91 ± 0.08	2.82 ± 0.10
Protein:albumin ratio	2.20 ± 0.06	2.26 ± 0.13	2.20 ± 0.03	2.21 ± 0.04	2.21 ± 0.04	2.21 ± 0.04
Sodium (mmol/L)	141.93 ± 1.44	141.28 ± 0.22	141.74 ± 1.75	140.76 ± 0.34	140.86 ± 2.68	142.56 ± 4.87
Potassium (mmol/L)	4.28 ± 0.15	4.50 ± 0.42	4.58 ± 0.35	4.86 ± 0.75	4.20 ± 0.10	4.38 ± 0.28

Note: The number of animals per group (*n*) is 5. Data are presented as mean ± SD; statistical analysis was performed using one-way ANOVA followed by Dunnett's multiple comparison test.

*p-value < 0.05 (control [G1] vs. individual treatments [G2–G4], and G1-R vs. G4-R).

3 | RESULTS

3.1 | Phytochemical analysis

BPGrit is an Ayurvedic formulation containing phytochemicals known for their antioxidant, anti-hyperlipidemia, and anti-hypertension properties. HPLC analysis of the BPGrit powder hydromethanolic solution showed the presence of 1.95 μg/mg gallic acid (13.69 min), 0.53 μg/mg ellagic acid (38.97 min), 0.09 μg/mg coumarin (42.69 min), 0.01 μg/mg cinnamic acid (44.78 min), 0.08 μg/mg guggulsterone E (52.59 min), and 0.05 μg/mg guggulsterone Z (53.83 min) (Figure 1).

3.2 | Morbidity, mortality, ophthalmoscopy analysis, and clinical signs

Repeated sub-acute oral gavage of BPGrit (100, 300, and 1000 mg/kg body weight/day) in Sprague–Dawley rats did not cause mortality, morbidity, abnormal clinical signs, or ophthalmic defects in both the sexes of treatment groups (Table S2). The G1-R and G4-R rats also did not show any incidence of mortality, morbidity, or abnormal clinical and ophthalmic signs over the recovery period of 2 weeks (Table S2). Intermittent detailed clinical assessment of the animals showed all of them to be in good health throughout the study period (Table S3).

3.3 | Body weight gain and feeding habits

Body weights and food consumption of the male and female Sprague–Dawley rats were measured once a week throughout the study period (Figure 2). On Day 28, the male animals showed body weights of 342.47 ± 12.64 g (G1), 341.48 ± 38 g (G2), 355.48 ± 24.05 g (G3), and 357.39 ± 8.8 g (G4), and the female animals showed body weights of 237.7 ± 21.78 g (G1), 230.5 ± 9.00 g (G2), 221.91 ± 16.64 g (G3), and 240.6 ± 14.49 g (G4), indicating no significant change following treatment with varying concentrations of BPGrit (Figure 2A,B). On Day 42, the recovery phase male animals

TABLE 5 Organ weight relative to terminal body weight (g) of Sprague–Dawley rats.

		Groups					
		G1 (vehicle control)	G2 (100 mg/kg body weight/day)	G3 (300 mg/kg body weight/day)	G4 (1000 mg/kg body weight/day)	G1-R (vehicle control)	G4-R (1000 mg/kg body weight/day)
Fasting body weight (g)	Male	317.42 ± 15.31	314.22 ± 34.13	331.33 ± 21.21	332.82 ± 10.21	391.48 ± 33.47	365.38 ± 23.36
	Female	219.88 ± 19.61	211.57 ± 5.99	207.97 ± 14.40	223.94 ± 15.51	221.04 ± 13.93	226.33 ± 14.27
Liver	Male	4.048 ± 0.174	4.020 ± 0.391	3.913 ± 0.335	4.287 ± 0.112	3.638 ± 0.229	3.889 ± 0.254
	Female	3.914 ± 0.242	3.823 ± 0.136	3.604 ± 0.244	3.772 ± 0.294	3.440 ± 0.384	3.528 ± 0.224
Kidneys	Male	0.863 ± 0.057	0.790 ± 0.070	0.824 ± 0.048	0.786 ± 0.044	0.738 ± 0.033	0.775 ± 0.066
	Female	0.807 ± 0.061	0.728 ± 0.043	0.811 ± 0.056	0.801 ± 0.036	0.703 ± 0.053	0.683 ± 0.070
Adrenals	Male	0.014 ± 0.003	0.0134 ± 0.004	0.014 ± 0.002	0.013 ± 0.004	0.015 ± 0.003	0.016 ± 0.003
	Female	0.033 ± 0.002	0.035 ± 0.005	0.033 ± 0.006	0.036 ± 0.003	0.032 ± 0.006	0.034 ± 0.003
Spleen	Male	0.214 ± 0.031	0.209 ± 0.014	0.216 ± 0.014	0.219 ± 0.019	0.201 ± 0.024	0.214 ± 0.029
	Female	0.225 ± 0.022	0.256 ± 0.029	0.219 ± 0.028	0.244 ± 0.012	0.241 ± 0.038	0.220 ± 0.025
Brain	Male	0.622 ± 0.025	0.627 ± 0.049	0.607 ± 0.061	0.592 ± 0.022	0.513 ± 0.032	0.533 ± 0.035
	Female	0.831 ± 0.075	0.848 ± 0.045	0.937 ± 0.056*	0.818 ± 0.030	0.854 ± 0.034	0.824 ± 0.054
Heart	Male	0.356 ± 0.019	0.359 ± 0.037	0.343 ± 0.008	0.343 ± 0.021	0.312 ± 0.012	0.358 ± 0.065
	Female	0.371 ± 0.025	0.377 ± 0.028	0.363 ± 0.010	0.358 ± 0.016	0.337 ± 0.024	0.325 ± 0.012
Thymus	Male	0.112 ± 0.010	0.115 ± 0.016	0.104 ± 0.007	0.103 ± 0.007	0.080 ± 0.006	0.099 ± 0.020
	Female	0.191 ± 0.012	0.196 ± 0.009	0.205 ± 0.031	0.238 ± 0.029*	0.170 ± 0.025	0.190 ± 0.018
Testes	Male	1.012 ± 0.080	1.031 ± 0.067	1.063 ± 0.103	1.101 ± 0.034	0.952 ± 0.047	0.986 ± 0.075
Ovaries	Female	0.060 ± 0.008	0.069 ± 0.018	0.068 ± 0.006	0.060 ± 0.010	0.062 ± 0.013	0.055 ± 0.015
Epididymides	Male	0.266 ± 0.028	0.269 ± 0.029	0.253 ± 0.028	0.274 ± 0.022	0.287 ± 0.020	0.294 ± 0.008
Uterus + cervix	Female	0.298 ± 0.071	0.367 ± 0.154	0.338 ± 0.084	0.353 ± 0.160	0.459 ± 0.073	0.384 ± 0.047
Prostate + seminal vesicle	Male	0.624 ± 0.172	0.620 ± 0.046	0.602 ± 0.140	0.595 ± 0.024	0.658 ± 0.079	0.612 ± 0.104

Note: The number of animals per group (*n*) is 5. Data are presented as mean ± SD; statistical analysis was performed using one-way ANOVA followed by Dunnett's multiple comparison test.

*p-value < 0.05 (control [G1] vs. individual treatments [G2–G4], and G1-R vs. G4-R).

showed body weights of 236.72 ± 14.9 g (G1-R) and 242.12 ± 14.93 g (G4-R) (Figure 2A,B). The recovery phase female animals showed body weights of 420.60 ± 40.81 g (G1-R) and 395.23 ± 27.42 g (G4-R) (Figure 2A,B). BPGrit-treated and recovery group (both sexes) animals also did not show any significant change in their feeding habits throughout the study period (Figure 2C,D). On average, the animals consumed 171.86 ± 10.43 mg/kg body weight/day feed (Figure 2C,D). Thus, the BPGrit-treated animals did not show any immediate or latent effect on their weight or feeding habits.

3.4 | Behavioral, neurologic, and autonomic functional analysis

No visible changes in the total movements made by the animals were observed in the BPGrit-treated and recovery group animals during the fourth and sixth weeks, compared with their respective control animals (Tables S3 and S4). When the relaxed animal moves around, contacting relevant stimuli (rearing), it represents its relaxed and exploratory behavior (McGregor et al., 2020). In our study, following treatment with the BPGrit medicine, a significant (p-value < 0.05) increase in the number of rearings was observed in the G4 male rats (36.40 ± 2.79) compared with the G1 animals (27.60 ± 7.06), indicating a more exploratory behavior (Table S3). However, the rearing movement was found to be temporary and absent in recovery (G4-R) group male animals, indicating it to be immediately associated with the BPGrit treatment. Furthermore, no change was observed in behavioral (posture, ease of removal, ease of handling, alertness, mobility, fur appearance, approach response, touch response, and startle response), neurological (convulsions and tremors, muscle tone, gait, stereotypic movements, tail pinch response, eye blink response, landing foot splay, and air righting reflex), or autonomic (palpebral closure, lacrimation, salivation, eye color, pupil function in response to light, skin color, piloerection, ease and rate of respiration, and fecal consistency) functioning in the male and female animals following exposure to BPGrit (Tables S3 and S4).

3.5 | Urine analysis

No blood was detected in the urine collected from the male and female treatment and recovery animal groups (Table S5). Urobilinogen, ketone bodies, glucose, protein, nitrite, leukocytes, clarity, color, specific gravity, and pH levels were found to be random in the treatment and recovery groups of male and female animals, demonstrating

TABLE 6 Summary of histopathological observations in Sprague-Dawley rats.

	Male						Female					
Dose (mg/kg body weight/day)	0	100	300	100	0	1000	0	100	300	1000	0	1000
Group	G1	G2	G3	G4	G1-R	G4-R	G1	G2	G3	G4	G1-R	G4-R
Number of animals examined	5	5	5	5	5	5	5	5	5	5	5	5
Liver												
Extramedullary hematopoiesis												
Multifocal, minimal	1	NA	NA	1	NA	NA	1	NA	NA	1	NA	NA
Bile duct hyperplasia												
Focal, minimal	1	NA	NA	1	NA	NA	1	NA	NA	1	NA	NA
Kidney												
Tubular basophilia												
Focal, mild	1	0	0	1	0	1	1	0	1	1	0	0
MNC infiltration												
Focal, minimal	1	1	0	1	0	0	1	0	0	1	1	0
Lungs												
Thickened alveolar septa												
Patchy, minimal	2	NA	NA	2	NA	NA	1	NA	NA	2	NA	NA
Proliferation of lymphoid tissue												
Focal, minimal	1	NA	NA	1	NA	NA	0	NA	NA	0	NA	NA
Ileum												
MNC infiltration												
Focal, minimal	0	NA	NA	0	NA	NA	0	NA	NA	1	NA	NA

Note: The number of animals per group (n) is 5.
Abbreviations: MNC, mononuclear cell; NA, not applicable.

no link to the BPGrit treatments (Table S5). Only one recovery phase female G4-R rat had 1 mg/dL of bilirubin in urine, indicating it to be an arbitrary incident (Table S5).

3.6 | Hematological analysis

The G4 male rats had a statistically significant (p-value < 0.05) decline in their blood lymphocyte percent (83.40 ± 1.34%) compared with the G1 rats (88.00 ± 1.41%), along with a statistically significant (p-value < 0.05) reduction in their prothrombin test time (G4: 16.48 ± 1.48 s; G1: 18.96 ± 0.92 s) (Table 1). The recovery phase male G4-R rats had a significantly higher white blood cell count (11.16 ± 0.78 × 10^3/μL) than the G1-R control (9.58 ± 0.94 × 10^3/μL) rats (Table 1). All other hematological parameters in both male and female Sprague–Dawley rats showed no significant alteration (Tables 1 and 2).

3.7 | Clinical chemistry analysis

A significant (p-value < 0.05) reduction in the ALT was observed in the 300 and 1000 mg/kg body weight/day BPGrit-treated male rats (G3: 41.38 ± 2.90 U/L; G4: 41.30 ± 1.80 U/L), compared with the control animals (G1: 50.64 ± 1.92 U/L) (Table 3). The recovery phase male G4-R animals had a statistically significant (p-value < 0.05) reduction in creatinine levels (0.35 ± 0.05 mg/dL), as well as an increase in blood glucose (153.18 ± 16.37 mg/dL), compared with the G1-R (creatinine: 0.42 ± 0.03 mg/dL; glucose: 1125 ± 16.37 mg/dL) group animals (Table 3). A statistically significant (p-value < 0.05) reduction in total bilirubin level was seen in the BPGrit-treated recovery phase female G4-R (0.08 ± 0.01 mg/dL) rats compared with their control (G1-R: 0.11 ± 0.02 mg/dL) counterparts (Table 4). Other biochemical markers tested in the animal groups showed no alterations following BPGrit treatments.

3.8 | Relative organ weight and histopathological analysis

BPGrit treatment had no effect on male or female animal groups' organ weights relative to their terminal body weights (Table 5). A thorough gross and histological study showed random anomalies present in the various animal organs, which were found to be unrelated to the BPGrit treatment (Table 6 and Figures 3, 4, and 5). The histopathological observations correlated well with the other observations made, confirming that BPGrit had no toxicologically important findings up to the tested dose of 1000 mg/kg body weight/day (Figures 3, 4, and 5).

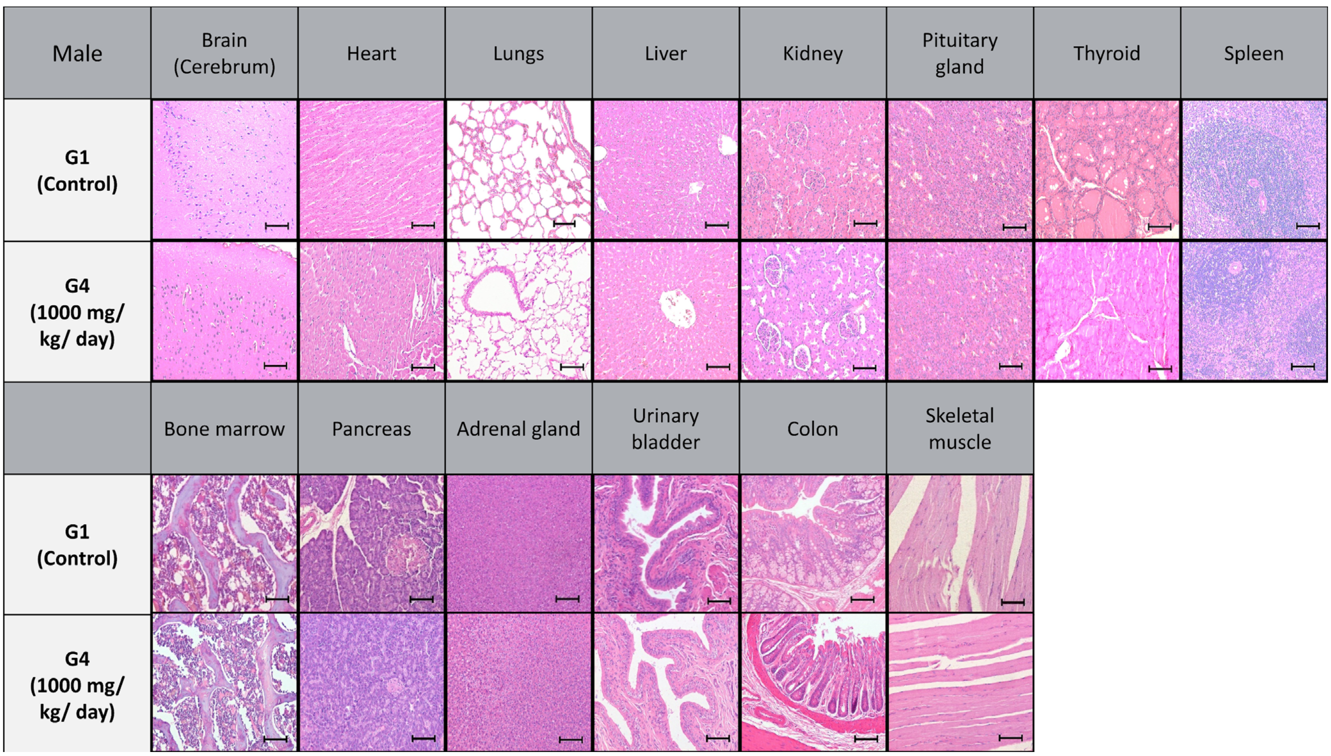

FIGURE 3 Histopathological analysis in male rats: The effect of 28 days repeated oral gavage of BPGrit was analyzed on different organs in male rats, like the brain (cerebrum region), heart, lungs, liver, kidney, pituitary gland, thyroid, spleen, bone marrow, pancreas, adrenal gland, urinary bladder, colon, and skeletal muscles. Representative histopathological micrographs represent male animals receiving vehicle control (G1) and 1000 mg/kg body weight/day (G4) of BPGrit over a period of 28 days.

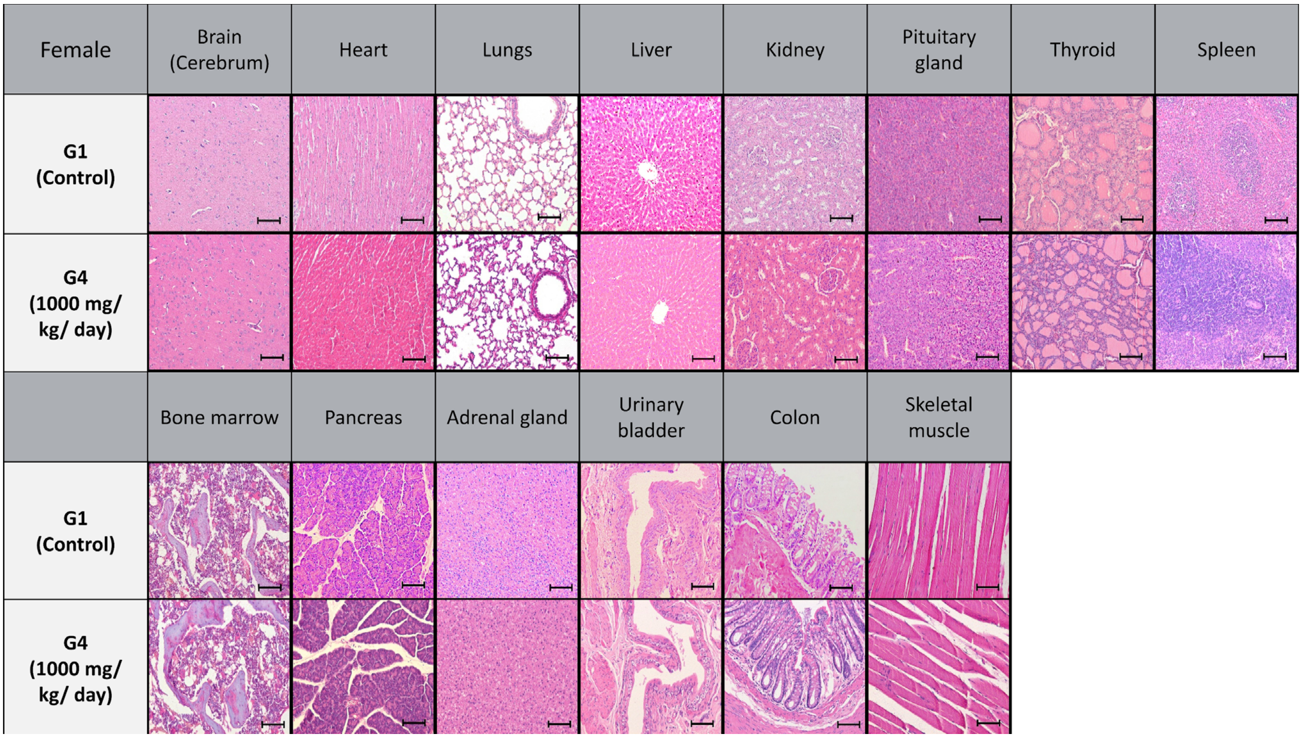

FIGURE 4 Histopathological analysis in female rats: The effect of 28 days repeated oral gavage of BPGrit was analyzed on different organs in female rats, like the brain (cerebrum region), heart, lungs, liver, kidney, pituitary gland, thyroid, spleen, bone marrow, pancreas, adrenal gland, urinary bladder, colon, and skeletal muscles. Representative histopathological micrographs represent female animals receiving vehicle control (G1) and 1000 mg/kg body weight/day (G4) of BPGrit over a period of 28 days.

4 | DISCUSSION

The global landscape of allopathic and herbal medicine is constantly evolving, driven by the increasing demand for effective treatments and therapies. As classical medicines are being explored through the lenses of modern scientific perspectives, it is imperative to ensure their biological safety. Herbal medicines play a major role in Indian society within the framework of Ayurveda (Prakash et al., 2017). The present study explored the phytochemical contents and toxicological evaluation of the polyherbal medicine BPGrit in male and female Sprague–Dawley rats. BPGrit contained phytochemicals like gallic acid, ellagic acid, coumarin, cinnamic acid, guggulsterone E, and guggulsterone Z originating from its herbal components: *C. mukul* (Hook. Ex Stocks) Engl., *T. arjuna* (Roxb. ex DC.) Wight & Arn., *T. terrestris* L., *P. granatum* L., *A. sativum* L., and *C. zeylanicum* Blume.

The present study explored the toxicity of BPGrit in male and female Sprague–Dawley rats over a 28-day treatment regimen, followed by a 14-day recovery phase. Repeated oral gavage of BPGrit showed no incidence of mortality, morbidity, or the development of aberrant clinical signs and ophthalmic symptoms up to the tested 1000 mg/kg body weight/day dosage throughout the study period. In an earlier study, gallic acid was found to have an NOAEL of 119 and 128 mg/kg body weight/day in male and female F344 rats, respectively (Niho et al., 2001). Similarly, ellagic acid was found to have an NOAEL of 3254 mg/kg body weight/day in female F344 rats (Tasaki et al., 2008). In an earlier study, treatment of male and female Sprague–Dawley rats with coumarin (male: 13 mg/kg body weight/day; female: 16 mg/kg body weight/day) caused significantly reduced survival following chronic exposure of 104 and 110 weeks, respectively (Carlton et al., 1996). A sub-chronic and chronic (90–180 days) toxicity study of *C. mukul* (Hook. ex Stocks) Engl. extracts containing 2.5%–7% w/w equivalent to 125–500 mg/kg body weight of guggulsterone E and guggulsterone Z exhibited no adverse health effects (National Toxicology, 2020). Hence, the quantified phytochemicals present in BPGrit were determined to be well within their reported NOAEL levels.

Sub-acute oral treatment of BPGrit in male and female Sprague–Dawley rats did not induce any change in the animals' body weight, eating habits, or behavior. In fact, after consuming the BPGrit medication, the rats exhibit more inquisitive behavior, as seen by an increase in rearing movement. This was absent in the recovery phase animals, indicating the behavior to be directly linked with the medicine intake. During the BPGrit treatment and recovery periods, no change in the rats' neurologic and autonomic functions was observed. Several studies have demonstrated that herbal components of the BPGrit, such as *C. mukul* (Hook. Ex Stocks) Engl., *T. arjuna* (Roxb. ex DC.) Wight & Arn., *T. terrestris* L., *P. granatum* L., *A. sativum* L., and *C. zeylanicum* Blume, have favorable health-oriented effects. The hydromethanolic extract of *C. mukul* (Hook. Ex Stocks) Engl. was reported to lower arterial pressure and enhance left ventricular end-diastolic pressure in a

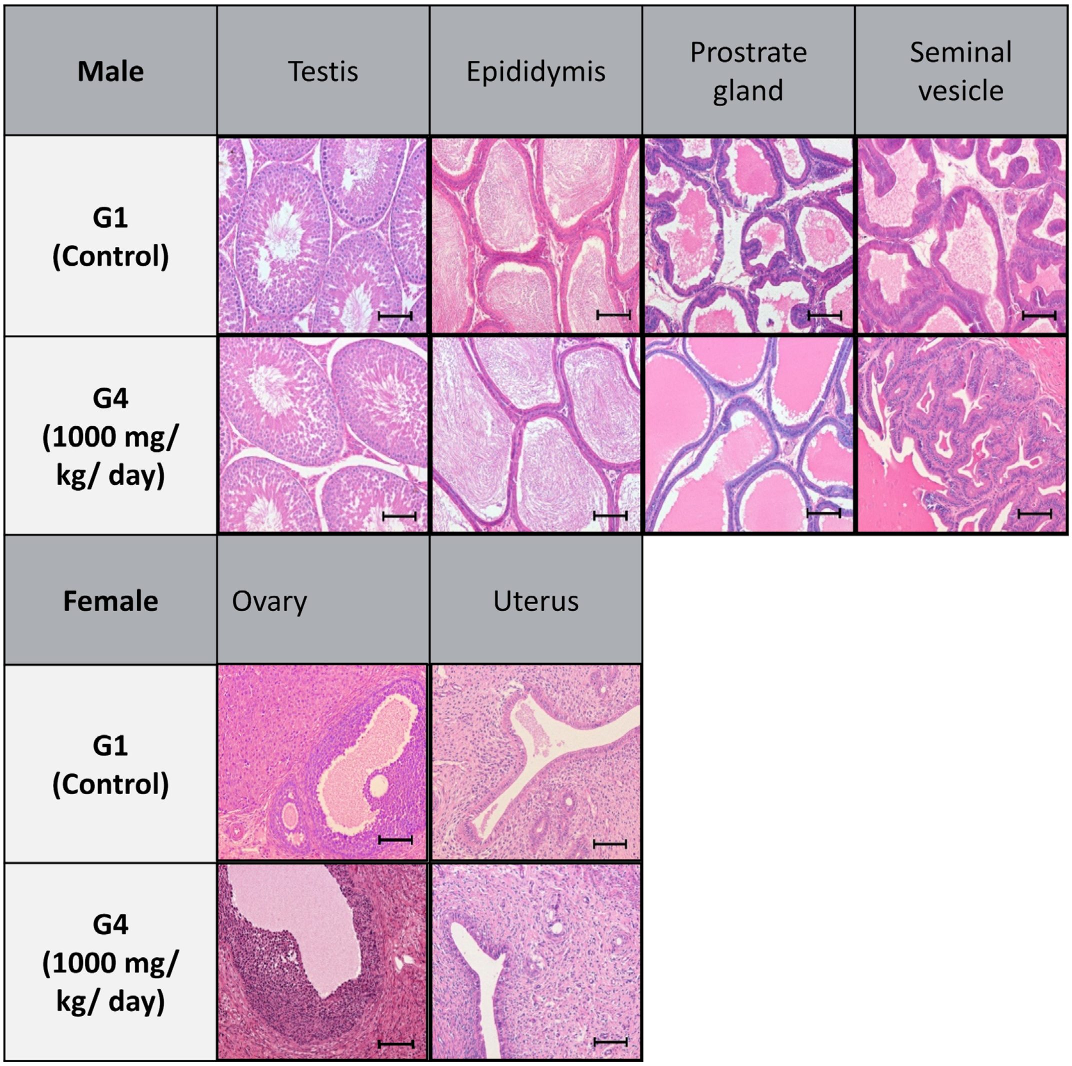

FIGURE 5 Histopathological analysis in male and female rats' reproductive organs: The effect of 28 days repeated oral gavage of BPGrit was analyzed on different organs in male and female rats' reproductive organs, like testis, epididymis, prostate gland, seminal vesicle, ovary, and uterus. Representative histopathological micrographs represent male and female animals receiving vehicle control (G1) and 1000 mg/kg body weight/day (G4) of BPGrit over a period of 28 days.

rat ischemia model and did not induce any toxicity in rats, dogs, or monkeys up to the studied dose of 500 mg/kg body weight (National Toxicology Program [NTP], 2020; Ojha et al., 2008). Suganthy et al. (2018) reported a no-adverse-effect oral dose of 1000 mg/kg body weight for *T. arjuna* Roxb. (ex DC.) Wright & Arn. bark methanolic extract in Swiss albino mice following OECD 420 (acute toxicity) and OECD 407 (sub-acute toxicity) guidelines. *T. terrestris* L. was shown to be non-toxic in female Wistar rats at tested dosages of 1000–5000 mg/kg body weight under sub-acute and acute test conditions (Yasmeen et al., 2022). The LD_{50} for *P. granatum* L. in male and female OF-1 mice was reported at 731 mg/kg body weight (Vidal et al., 2003). The LD_{50} for *A. sativum* L. has been observed to be between 0.5 and 30 mL/kg body weight when administered to rats and mice through various routes (Dutta et al., 2021). In this study, the herbal components of BPGrit were found to be within their reported safe limits at the maximum tested dose in Sprague–Dawley rats.

Urine analytical parameters did not demonstrate any significant biochemical changes following BPGrit exposure in the male and female rats during the treatment and recovery periods. In the hematological analysis, minor alterations were observed in lymphocyte and white blood populations and prothrombin test time during BPGrit treatment, but they were found to be well within their historically reported reference range at the study site. Their normal laboratory background limits have also been validated by Delwatta et al. (2018) at the Faculty of Medicine, University of Colombo, Sri Lanka, and Wohlauer et al. (2011) at the Department of Surgery, University of Colorado Denver. Hence, these findings were found to be coincidental and non-correlated with BPGrit treatment in male and female animals.

Changes observed in the blood serum levels of ALT, glucose, and bilirubin in rats orally treated with medium and high dosages of BPGrit were within the normal laboratory range reported in other studies (Delwatta et al., 2018) and historical reference ranges reported within the study center. The results were further validated by the histopathological observations made in the BPGrit-treated male and female rats, indicating normal liver tissue. Hence, the changes were coincidental and not associated with the BPGrit treatment. Modulation in kidney function can be validated through the measurement of serum and urine creatinine, albumin, urea, sodium, and potassium levels (Gounden et al., 2023). Kidney injuries have been associated with an increase in serum creatinine levels, ALP, and a reduction in sodium and potassium levels (Barbosa et al., 2020; Briguori et al., 2018; Damera et al., 2011; Kellum et al., 2021). In our study, no changes were observed in the serum electrolyte levels in rats following treatment with BPGrit. Hypertension drugs have been found to affect renal functions, causing an increase in serum creatinine levels (Collard et al., 2018; Coresh et al., 2001). In contrast, BPGrit did not alter serum creatinine levels in the G4-R male rats beyond the historical reference range available at the study location, suggesting unaltered normal kidney functioning. The findings were consistent with the normal urine analytical and kidney histopathological results obtained from both the BPGrit treatment and recovery groups of male and female rats. Other vital glandular and reproductive organs showed normal tissue pathology in all the BPGrit treatment animal groups during the study and recovery phases. The NOAEL is a critical parameter that serves as a pivotal reference in guiding safe dosage levels for human consumption (OECD, 2008). Hence, based on the study outcome, the NOAEL for BPGrit was found to be 1000 mg/kg body weight/day based on the sub-acute toxicological study done in Sprague–Dawley rats.

5 | CONCLUSION

Sub-acute toxicological profiling of polyherbal Ayurvedic BPGrit medicine revealed it to be well tolerated in Sprague–Dawley rats after 28 days of repeated oral gavage in accordance with OECD 407 guidelines, followed by a 14-day recovery period. BPGrit had no influence on the male and female rats' behavioral, neurological, or autonomic functions throughout the study period, up to the highest tested dosage of 1000 mg/kg body weight/day. Similarly, no clinical, ophthalmological, or histological abnormalities were detected in the rats following BPGrit treatment until the maximum tested dose. The study established the NOAEL dose of BPGrit at >1000 mg/kg body weight/day.

AUTHOR CONTRIBUTIONS

Acharya Balkrishna: Conceptualization; planning; visualization; supervision; writing—review and editing. **Sandeep Sinha**: Methodology; investigation; data curation; formal analysis; writing—review and editing. **Kunal Bhattacharya**: Data curation; formal analysis; writing—original draft; visualization. **Anurag Varshney**: Writing—review and editing; project administration; conceptualization; visualization; supervision.

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CONFLICT OF INTEREST STATEMENT

Acharya Balkrishna is a trustee in the Divya Yog Mandir Trust, Haridwar, India, that governs Divya Pharmacy, Haridwar, and one of the founding promoters holding an honorary managerial position at Patanjali Ayurved Ltd., Haridwar, India. These organizations were not involved in any aspects of this study. All other authors, Sandeep Sinha, Kunal Bhattacharya, and Anurag Varshney, are employed at the Patanjali Research Institute, which is governed by the Patanjali Research Foundation Trust (PRFT), Haridwar, Uttarakhand, India, a not-for-profit organization. In addition, Anurag Varshney is an adjunct professor in the Department of Allied and Applied Sciences, University of Patanjali, Haridwar, India, and in the Special Centre for Systems Medicine, Jawaharlal Nehru University, New Delhi, India. All other authors declare no competing interests.

DATA AVAILABILITY STATEMENT

The datasets generated during and/or analyzed during the current study are available from the corresponding author on request.

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